

1.

- VAEs are optimized to maximizing the variational lower bound, assuming each pixel is drawn from an independent Gaussian. The decoder outputs a deterministic mean of the predicted distribution, which can act like an average, contributing to blur. GANs are trained with an objective that directly rewards outputs that fool a discriminator judging realism, so if the generator produces an image that is too blur, the discriminator can tell it which leads to higher penalty.

2.

It learns to distinguish between real images and fake images generated by the generator. It outputs a probability indicating how real an image looks. The generator learns from the discriminator's feedback. When the discriminator says a generated image is fake, the generator updates its parameters to try to fool the discriminator so it can generate more realistic images.